BASE CHANGE OF A DUALIZING COMPLEX ALONG A FLAT MAP WITH CONTROLLED FIBRES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The base change statement for internal homs along flat maps. Invoked at the start of the counted proof (source lines 4295-4301) as a cross-chapter pointer. Transfers the internal hom along the base change.
- The criterion for a bounded complex of coherent modules to have finite injective dimension over a Noetherian ring. Invoked inside the counted proof (source lines 4302-4308) as a cross-chapter pointer. Gives finite injective dimension of the base change.

Source reference. Original reference more-algebra-lemma-base-change-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2719–2719. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-base-change-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-finite-injective-dimension-Noetherian-radical: sources/stacks-dualizing/pinned-original-source.tex, source lines 4307–4307. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-finite-injective-dimension-Noetherian-radical, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $A \to B$ be a flat map of Noetherian rings. Let $I \subset A$ be an ideal such that $A/I = B/IB$ and such that IB is contained in the Jacobson radical of B. Let $\omega_A^\bullet$ be a dualizing complex. Then $\omega_A^\bullet \otimes_A B$ is a dualizing complex for B.*

Proof. It is clear that $\omega_A^\bullet \otimes_A B$ is in $D^b_{Coh}(B)$. By More on Algebra, Lemma more-algebra-lemma-base-change-RHom we see that

$$R\operatorname{Hom}_B(K \otimes_A B, \omega_A^\bullet \otimes_A B) = R\operatorname{Hom}_A(K, \omega_A^\bullet) \otimes_A B$$

for any $K \in D^b_{Coh}(A)$. For any ideal $IB \subset J \subset B$ there is a unique ideal $I \subset J' \subset A$ such that $A/J' \otimes_A B = B/J$. Thus $\omega_A^\bullet \otimes_A B$ has finite injective dimension by More on Algebra, Lemma more-algebra-lemma-finite-injective-dimension-Noetherian-radical. Finally, we also have

$$R\operatorname{Hom}_B(\omega_A^\bullet \otimes_A B, \omega_A^\bullet \otimes_A B) = R\operatorname{Hom}_A(\omega_A^\bullet, \omega_A^\bullet) \otimes_A B = A \otimes_A B = B$$

as desired. □